Blue outlines around figures indicate images taken by the Artemis II crew or the Orion vehicle cameras. On occasion, the crew used clock-direction vernacular to describe positions of features. Orion pointing was nose-to-lunar-north (i.e., the docking-hatch window faced the approximate same direction as the lunar north pole). Crew orientation within the window, however, varied over time and between crew members, ranging from aligned with Orion (i.e., viewing the Moon as north-up) to anti-aligned (viewing the Moon as south-up). Therefore, it is important to interpret crew descriptions using the contemporaneous crew images.

5.1 Impact Flash Observations

The present-day cis-lunar environment is dynamic, with impacts from micrometeoroids, which are naturally occurring fragments of comets, asteroids, and planetary impact debris that travel through space. When these meteoroids strike the lunar surface, the kinetic energy of the impact creates seismic waves (Moser et al., 2015; Madiedo et al., 2019).

These impact events can be visible as a bright flash on the lunar surface and hence are observable from Earth and in cislunar space (Ortiz et al., 2000). In at least three instances, Apollo astronauts recorded observing impact flashes from orbit: one impact flash during Apollo 16 and two flashes during Apollo 17. According to NASA's Meteoroid Environment Office (MEO), at the time of this writing 443 impact flashes have been observed by Earth-based telescopes through a variety of observing campaigns, and NASA's Lunar Reconnaissance Orbiter (LRO) spacecraft has documented multiple new impact craters that have formed since its launch in 2009 (Moser et al., 2015; Sheward et al., 2022, 2025; Rizos et al., 2026).

Given previous observations of impact flashes during Apollo (NASA, 1972a, 1972b), plus ground-based observations and LRO datasets (e.g., Suggs et al., 2014; Moser et al., 2015; Sheward et al., 2022, 2025; Rizos et al., 2026), the LST strategically planned observation time for impact flash surveys, both by crew (**Appendix 1**) and by the Orion vehicle exterior cameras. More specifically, the crew had dedicated time scheduled to observe impact flashes as a specific LTP target and during the eclipse period of the flyby. The eclipse period enabled observations at a time when the Sun was eclipsed behind the Moon. This provided dark viewing conditions with the Moon not directly illuminated by sunlight (only partially lit by earthshine) and removed any direct sunlight within the crew's field of view (FOV), which aided with dark adaption for the crew's eyes to adjust and be able to see fainter phenomena. Furthermore, the crew were trained to report any impact flashes observed at any point during the mission, even outside of scheduled surveys.

Additionally, a coordinated Earth-based observation campaign was also undertaken, including the MEO at NASA's Marshall Space Flight Center through the Lunar Impact Flash Monitoring Program and complemented by an SMD citizen science campaign to increase public engagement and observing opportunities (**Section 6.2**). Earth-based observations were optimized to collect data near the times of the Artemis II crew observations, though this was not always possible given that ground-based observations require views of shadowed (non-illuminated) lunar terrain, which were not always present from the vantage point of Earth during the crew's lunar flyby.

5.1.1 Characterizing Impact Events

Observations of impact flashes (or a lack thereof) during dedicated observation periods help characterize the modern impact environment and the scale of what may be observable from cis-lunar space. Of particular interest are constraining the meteoroid size and frequency distributions, as well as the timing, location, and brightness of individual flashes. The frequency distribution of impact flashes across the lunar surface can help constrain the impactor flux as a function of geographic location on the Moon (Ortiz et al., 2006). The frequency of flash observations, correlated with the presence (or absence) of known meteor showers during the observing periods, can provide insight into the sources and rates of the flashes (Madiedo et al., 2015; Ortiz et al., 2002). The timing and location of flashes

also enable follow-up studies to possibly image and characterize resulting craters and to investigate impact-crater processes. In addition, analysis of the observational data can reveal whether and how future measurements might constrain the seismic source and timing of impacts, especially once ground-based geophysical instrumentation is on the lunar surface. The brightness (or intensity) of the flash itself can be correlated with the size of the impactor and resultant energy of the impact event (Bellot Rubio et al., 1998). For Artemis II, the strongest impact-flash-related dataset is the number of observed impact flashes (i.e., their frequency distribution).

The Artemis II impact-flash observation campaign was designed to inform techniques for future observations and guide subsequent science investigations. For example, the statistics of impact-flash detections made by the crew, correlated with corresponding observing time, provide data on the detection frequency and probability for varying observing durations on future missions. These results can support the planning of observing surveys on crewed missions, as well as the selection of optimal detector systems for future flight missions. Additionally, the observing assets on Artemis II may constrain the expected frequency and magnitude of future impact flashes, thereby informing detection limits for future observations and highlighting the potential value of additional observing techniques, such as a dedicated automated flash-detection camera with increased sensitivity and/or resolution (**Section 9.7**). Overall, impact flash observations will contribute to future Artemis mission planning by improving understanding of the cis-lunar space environment and present-day meteoroid activity.

5.1.2 Impact Flash Observations

Five distinct impact flashes were reported by the crew during the period of the solar eclipse as seen from the vantage point of the Orion spacecraft. The crew possibly observed several additional impact flashes but did not formally report them. Crew members reported flashes in the crew PCD audio files, through O/E audio communication live during the flyby, and via an annotated image that was prepared by the crew and downlinked to the LST after the flyby on which the crew marked their approximation of where the five confirmed flashes were located on the lunar surface (**Figure 5.1**). Several of the flashes were observed simultaneously by multiple crew members, cross validating the observations as true impact-flash detections.

All definitive flashes observed by the crew were seen during the lunar eclipse period of the lunar flyby, and no flashes were associated with any color. All the impact flashes were observed on the non-illuminated portion of the Moon, and no flashes were reported in earthshine during the eclipse. No flashes were observed when any portion of the Moon was illuminated by the Sun; therefore, these flashes are likely the faintest ever observed by human eyes, suggesting small impactors on the order of several cm in diameter. Artemis II flew during a low meteoroid flux time of year outside of any major annual meteoroid streams, and thus these observations place a lower bound on the impact flux at the lunar surface.

The first impact flash was reported by R. Wiseman:

> I just saw a flash! 100% I just saw flash! 100% it was on the like if you go right to the center point, it was at the 7 or 8 o'clock. (R. Wiseman, camera, PCD crew audio, art002a000052, 00:41:01 UTC)

Shortly after, R. Wiseman and J. Hansen reported observing the same impact flash:

R. Wiseman	Oh, I just saw another meteorite. 100%. Just saw a meteorite.
J. Hansen	Did you see it in the, was it lower?
R. Wiseman	Yeah. Yup.
J. Hansen	Like a third from the bottom, up from the bottom?
R. Wiseman	Yeah.
J. Hansen	Yup.
J. Hansen	But it's a pinpoint of light.
R. Wiseman	Yup.
J. Hansen	It's not like a meteor, you didn't see anything else, it's just like a bloop.
R. Wiseman	Yeah.
J. Hansen	So, Reid and I both saw the same thing there. We saw an impact flash. It wasn't very big. For me it was a pinprick. It was about a third of the way up from the bottom of the Moon for us.

R. Wiseman, camera, and J. Hansen, unaided eye, PCD crew audio, art002a000052, 00:51:15 UTC)

During the post-flyby crew conference between the crew and the Science Officer on Flight Day 07, the crew described that these observations were reported as impact flashes because they were distinct from other transient phenomena, such as cosmic ray hits to the eye or spacecraft glint. During the conference, the crew also reported that more flashes may have been observed than were recorded. The crew intentionally did not include any ambiguous flashes in their annotation (**Figure 5.1**), thus lending more credence to the validity of the reported detections.

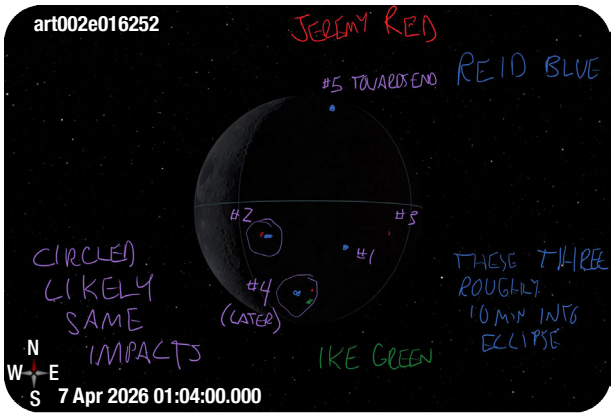


Figure 5.1: Impact Flash Annotation. Annotation art002e016252 made in-flight by the Artemis II crew showing general locations of reported impact flashes observed by R. Wiseman (blue), J. Hansen (red), and V. Glover (green). Purple was used for annotation and not representative of a specific crew member.

The MEO at Marshall assessed a SAW Cam 3 video that recorded for ~2 minutes during the lunar eclipse at its highest frame-rate setting (30 frames/second). No impact flashes were detected in this dataset, though the video covered a limited time period. Additional details on this SAW Cam 3 video are available in the *Artemis II Lunar Science Data User Guide*.

The LST also collaborated with the LRO team and determined that potential follow-up observations of new craters formed from the crew-observed impact-flash events are probably infeasible since the impact crater size is likely too small to be resolvable by the LRO Camera (LROC). Additionally, information related to impact flash locations from visual crew observations is insufficient for LRO targeting purposes.

5.2 Color Provinces and Albedo Variations

Crew data related to color provinces and albedo variations help address what humans can uniquely perceive at the Moon and provide information on lunar geologic features. Because none of the handheld or vehicle camera images were calibrated for this mission, analysis of color and photometric characteristics are necessarily qualitative. The crew's verbal descriptions of color serve as the primary science data for this theme. The accompanying crew images acquired during their verbal descriptions provide critical geologic context, including relative color and albedo properties (**Figure 5.2**), morphological

field deployments, simulations, and one-on-one tutorials. Throughout this process, the crew provided specific feedback about their training experience, informing the recommendations below for science and science operations training in future Artemis crews.

Among this feedback, the crew feedback emphasized the value of establishing a strong scientific foundation early in the training flow, alongside hands-on field experience that grounded classroom concepts in real geological processes. Field training in particular was identified as critical to learning how to observe and interpret geological features firsthand, in ways classroom instruction alone cannot replicate.

The crew also pointed to the importance of training that could flex to meet evolving mission needs, while still building skills in a structured, repeatable way. This included hands-on practice designed to progressively build both scientific "stamina" and final required skills, scenario-based exercises requiring collaborative problem-solving, and opportunities for real-time scientific feedback during simulations and one-on-one mentoring.

Finally, the crew underscored the value of self-directed learning tools that reinforced material between formal training sessions. Frequent, low-effort exposure to lunar science content with rapid turnaround feedback helped the crew retain critical knowledge over the course of a multi-year training flow. This self-directed and curiosity-driven training even extended into the mission itself. In addition to the LGR provided by the LST, the crew brought self-selected reference materials to the mission to continue reading and learning on their way to the Moon.

Taken together, this feedback provides the following recommendations to support future Artemis crew training:

Recommendation: Future Artemis crew training should maintain an early, introductory learning experience that provides foundational and contextual information about the Moon and the Artemis campaigns and should continue to prioritize field training as a critical tool for learning geological processes.

Recommendation: Future Artemis crew training should develop flexible training modules that address just-in-time training needs for the crew and for science; build training opportunities into simulations, with scientists providing real-time feedback: assign geology mentors to individual crew members to extend this real-time mentoring and feedback availability beyond simulation events; and build and train to standardized procedures that can be practiced consistently across training environments.

Recommendation: Future Artemis crew training should create a robust, regularly updated portfolio of digital training products and short homework assignments that the crew and FCTs can access independently, providing frequent exposure to lunar science material with assessment and feedback delivered within one day to support knowledge retention.

9.4 In-Mission LST to Crew Communications

Real time, bidirectional communication between the LST and the crew proved to be a valuable asset throughout the mission, enabling the LST to respond to crew questions as they arose, request clarification during time-critical science events, and directly support science interpretation while data were being collected.

Several times during the lunar flyby, the crew requested scientific information from the LST. For example, the crew asked for assistance with identifying surface features they were observing on the Moon in real time or determining relative ages of observed targets. Because the LST staffed in the SER had direct communication with the Science Officer, and because the Science Officer had direct communication with the crew, these requests received immediate, in-mission responses.

During the solar eclipse period, the crew reported seeing several impact flashes. With direct voice communication, the LST was able to immediately solicit information on the timing and locations of the flashes, allowing the team to determine with greater confidence the number of flashes observed, which flashes were observed by multiple crew members, and where on the lunar surface the flashes were located. This additional data collection that was enabled through real-time communication between the LST and crew at the Moon proved critical for subsequent scientific analysis of these impact flash data.

an STM. This will enable efficient mission science planning when the exact mission profile may not be known until a short time before launch or until after launch.

Recommendation: Future Artemis missions should include extensive pre-mission preparation in the form of mission simulations focused on planning and doing science in the context of a wide range of plausible mission scenarios. This will best exercise the expertise, confidence, and flexibility needed in a well-prepared science team and also result in STM-responsive products that provide the science team with early progress on actual mission planning and execution.

9.6 Calibration of Cameras

Camera calibration will be essential for future missions to independently contextualize crew observations and enable quantitative analyses. Within the Artemis II mission architecture, the LST was not afforded the opportunity to select instrumentation or specify calibration needs in support of science. As discussed in **Section 3.2.2**, the cameras used during Artemis II were COTS systems with only partial pre-flight characterization, and no geometric, radiometric, or spectral calibration were completed on any of the included imaging systems.

To enable quantitative use of images in future missions, it will be important to specify the minimum required calibration level needed to support science objectives. Camera characterization and calibration are needed to understand internal parameters (e.g., sensor response) and external factors (e.g., window effects; **Section 7.2**), as both influence image quality and the interpretation of color, brightness, and shape. Additionally, the performance of manifested camera systems, including handheld, vehicle-mounted, and EVA tools, should be characterized in the unique lighting, thermal, and viewing environments in which they will be utilized.

Recommendation: Future Artemis missions should define minimum camera calibration requirements and characterize camera performance inside and outside the spacecraft to enable quantitative and independently verified crew observations.

9.7 Instrumentation for Enhanced Science Return

Artemis II highlighted several areas where dedicated science instrumentation would substantially improve the quality, interpretability, and diversity of lunar observations. Many of the lunar science objectives, particularly those associated with the characterization of impact flashes, color and albedo, photometry, dust detection, and deep-sky imaging, would have benefited from instruments specifically designed to address the unique observational challenges of cislunar space. Several illustrative examples of instruments for enhanced science return are outlined below. Missions to the surface of the Moon may have different instrument needs based on science objectives and the nature of in situ versus remote measurements (e.g., surface missions would benefit from strategically selected handheld field instruments for crew members).

Although the crew observed and described several impact flashes during the solar eclipse, a dedicated impact-flash sensor that is capable of high-frame-rate, high-dynamic-range imaging would allow systematic monitoring of impact events, enable quantitative flash brightness measurements, improve temporal precision and location identification, and enable direct comparison with human observations.

Calibrated and more capable imaging and/or spectrometer systems would improve collection of color and albedo data. The Artemis II crew frequently described subtle color variations across the lunar surface, but camera limitations prevented quantitative characterization. A calibrated multispectral camera and/or dedicated spectrometer system(s) would provide independent measurements of spectral variations and allow direct comparison between crew perception and instrument-derived reflectance and spectral properties, strengthening scientific interpretation of surface composition and photometric behavior.

Relatedly, a physical onboard color wheel would provide the crew with a stable, standardized reference for color hue and value, improving consistency and reproducibility in real-time descriptions of lunar

color and albedo. A color wheel proved to be a highly effective tool for supporting visual observations during the Apollo-Soyuz mission (El-Baz, 1977, 2011).

For investigations of the lunar dust environment, future missions would benefit from a dedicated low-light camera optimized for limb and horizon observations. Such a system would provide quantifiable measurements of extended faint sources and enable confident detection or non-detection of lofted dust.

Deep-sky imaging revealed optical constraints of general-purpose lenses. Coma and corner distortions were evident in wide-field images, which is expected for fast lenses not optimized for astronomical use. For future missions, particularly those spending longer durations in lunar orbit, a dedicated prime lens for deep-sky imaging would provide higher-quality stellar data and support investigations of zodiacal light and other faint phenomena.

As instrumentation needs will vary by science objectives and priorities, future development should be responsive to each mission profile, as well as community priorities, while maintaining compatibility with human-driven observation workflows.

Recommendation: Future Artemis missions should manifest dedicated science instruments that meaningfully support the science objectives of the mission, to include IVA instrumentation to support orbital observations similar to the Artemis II flyby observations, as well as EVA-specific instrumentation for use by astronauts on the lunar surface.

9.8 Window Design for Science Observations

For Artemis II, lunar images collected through Orion's windows were a cornerstone of the lunar science dataset. These images enabled cross-regional comparisons, supported interpretation and contextualization of the crew's verbal observations, and anchored the LST's ability to map observations to specific lunar features. These images were also affected by the vehicle window design, including window size, pane design, and optical quality, and thus window design is a critical component in enabling high-fidelity science observations.

The Artemis II window-quality DFTO results underscore the need to continue to characterize vehicle window designs. Windowpanes should be characterized individually and as an installed stack. This should be included for any updates to Orion's windows that may be implemented for future Artemis flight missions and also the windows for the Human Landing System vehicles that will deliver subsequent Artemis crews to the lunar surface. Lower quality windows and/or incomplete characterization of windows for imaging will have a detrimental effect on the science return from human observations and images, particularly regarding phenomena that Artemis II demonstrated humans are adept at observing and describing, such as impact flashes, nuanced color patterns, and regional comparisons.

Recommendation: Future Artemis missions should rigorously characterize and prioritize window optical performance of windowpanes and the assembled window stack (for Orion and Human Landing System vehicles) to maximize the quality of science return from crew observations and images.

9.9 Crew-Audio Recording Capabilities and Time Synchronization of Multi-Modal Datasets

This report highlights the importance of crew observations to the mission's science return. This highest-priority dataset was enabled on Artemis II via manual audio recordings captured by the crew on their PCDs. However, using PCDs for this critical scientific purpose introduced risks and challenges in several ways. First, the tablets were at significant risk of data loss due to radiation strikes. Without downlink capability, these critical data could have been permanently lost, as no backup solution existed. Second, each observation required the crew to manually start and stop the audio recordings on their PCDs. This added operational burden and increased the likelihood of missed activations or incomplete recordings, potentially compromising the dataset. Third, managing the PELTOR headsets, which were attached to the PCDs by lengthy cables floating in microgravity, proved cumbersome. By contrast, Apollo missions benefited from in-cabin recording devices that did not require any action from the

crew. These passive audio recordings have been used for decades as an indispensable historical record of the mission, as well as a valuable scientific record (e.g., recordings of the Apollo 11 crew making lunar observations in LOS before landing).

Further related to the value of audio recordings, the LST often relied on the sounds of camera shutter clicks as a useful audio cue for aligning multi-modal crew data (i.e., images, audio files, and annotations) with science intent during Artemis II (**Section 8.1**). This in-cabin cue worked well for IVA operations, where handheld images and voice-loop recordings were collected within the same acoustic environment and could be cross-referenced by listening for shutter sounds in the audio. For future surface missions, however, this approach will not always be viable. The sound of a Nikon D5 camera shutter or the simulated shutter sound of the Nikon Z9 will not reliably propagate to microphones embedded in EVA suits. Additionally, crew members on a surface EVA will not use PCDs for real-time annotations as IVA crews do, and will instead rely more heavily on verbal communication. As a result, correlating multi-modal science observations, including images, verbal descriptions, and other data streams during EVAs, will be even more dependent on precise and consistent timestamps across all devices.

Recommendation: Subsequent Artemis missions should incorporate a passive audio-recording capability to support the collection of crew observations. A continuous, timestamped onboard audio record would also provide an independent means of establishing the timing of activities not captured by other onboard systems, substantially improving post-mission reconstruction and interpretation of the sequence of mission events. This capability would be particularly beneficial during highly dynamic science mission phases, such as the Artemis II lunar flyby. Passive audio recording would reduce crew workload by eliminating the need to manage additional hardware during high-priority and time-critical periods. Additionally, passive audio would capture valuable scientific context by recording crew perception during mission phases where opportunistic or serendipitous science data is collected. This recommendation aligns with crew feedback in the crew perspectives section (**Section 2.0**).

Recommendation: Subsequent Artemis missions should maintain coordinated and correct timestamp synchronization across all science-relevant systems, including cameras, crew suits, PCDs, voice loops, any additional manifested science payloads, and the spacecraft, to ensure accurate alignment of multi-modal datasets during orbital and surface mission phases.